

Rohith Poola

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EDUCATION

Northeastern University, Boston MA

Master of Science in Robotics

Dec 2026

GPA: 4.0/4.0

Coursework: Robotics Sensing and Navigation, Mobile Robotics, Robot Mechanics and Control, Autonomous Field Robotics

National Institute of Technology Calicut, Kozhikode, India

Bachelor of Technology in Mechanical Engineering

May 2022

GPA: 3.84/4.0

SKILLS

Programming & Libraries : Python, C++, MATLAB, Simulink, JAX, CUDA, PyTorch, Bash

Tools & Platforms : ROS/ROS 2, MuJoCo, Drake, Isaac Sim, Linux, Docker, Git, Jetson

Proficiencies : Manipulation, Motion Planning & Control, Robot Learning, Sensor Fusion, SLAM

WORK EXPERIENCE

Medtronic (Surgical Robotics)

Jan 2026 – Aug 2026

Robotics Software Engineer (co-op)

Boston, MA

- Proposed and built a **learned-manipulation** pipeline with ACT and $\pi_{0.5}$ VLA policies, collecting surgeon-console demonstrations and deploying them through the surgical control interface for closed-loop randomized pick-and-place in simulation
- Characterized load-dependent **tendon transmission** loss and **backlash-induced** path deviation (up to 5 mm) through simulation and hardware experiments, quantifying nonidealities to inform compensation in a tendon-driven **surgical robot**
- Developed requirements-driven Simulink control software integrating **12+** subsystems, and commissioned it on a physical robotic arm through calibration, bus integration, V&V, and on-hardware debugging to enable **manual control** and **teleoperation**

Autonomy and Intelligence Lab – Northeastern University

Jan 2025 – Present

Research Assistant

Boston, MA

- Developed and hardware-validated **Contingency-MPPI**, a risk-aware planner for dynamic, uncertain environments that preserves a feasible escape route to safety at every step, with adversary detection and occlusion-based safe-zone extraction
- Engineered a **JAX**-accelerated ROS 2 **hardware-in-the-loop** planning pipeline, reducing cycle time 2× through efficient **RBR** resampling and enabling real-time closed-loop execution with an NMPC trajectory tracker ([link](#))

Capgemini (Automotive)

Aug 2022 – Aug 2024

Associate-II Software Engineer

Hyderabad, India

- Built **safety-critical** software that standardized **400+** channels of crash-test data from **100+** IMU, GPS, and pressure sensors and developed a custom visualization interface, establishing a reliable data pipeline for side-impact detection
- Trained and ensembled RNN, GRU, and Transformer models in PyTorch on the standardized pipeline, improving airbag trigger timing by **21 ms** over production logic in more than **90%** of cases

Robotics Research Centre – IIIT Hyderabad

May 2022 – Jul 2024

Research Assistant (part-time)

Hyderabad, India

- Designed underactuated tendon-driven and fully actuated **multi-fingered bionic hands** with custom Series Elastic Actuators for compliant force control, integrating them with an **xArm-6** via ROS ([link](#))
- Built a **Dagger-style** interactive **imitation-learning** loop for dexterous **in-hand manipulation** of Rubik's Cube, using kinesthetic teaching with motor torques disabled to correct failed hardware rollouts and retrain on aggregated corrective demonstrations

PROJECTS

Autonomous Harvesting Robot for Vertical Farms, Robotics Lab, NITC ([link](#))

- Engineered a 5-DOF mobile manipulator with a compliant grasp-and-cut end-effector for autonomous crop harvesting
- Developed the closed-loop **perception-to-manipulation** stack, combining custom-trained **YOLOv4** with stereo depth for **3D crop localization** and gravity-compensated **PID** joint control to execute approach, cutting, and binning

Multi-Robot SLAM, Field Robotics Lab, NEU ([link](#))

- Built and deployed a modular **C++/roscpp** multi-sensor acquisition stack for stereo, LiDAR, IMU, and RTK-GPS, synchronizing four asynchronous streams and sustaining full-rate capture across five heterogeneous robotic platforms
- Mapped two 7-floor GPS-denied buildings with **FAST-LIO**, merging per-robot maps into a drift-resistant **3D reconstruction**.

Off-Road Traversability Estimation, NAIL, NEU: Modeling traversability as a continuous **vehicle-terrain interaction** field that predicts slip, stability, and **failure risk** as functions of terrain, approach angle, and speed for joint path and velocity optimization

Additional Projects: [Obstacle-Aware Manipulation](#), [Non-Prehensile Manipulation with MPPI](#), [Acrobot Swing-up Control](#)

PATENTS AND PUBLICATIONS

- Rohith Poola et al., “Mobile Robotic Manipulator Based Harvesting Inside Vertical Farms”, In: 2023 International Conference of The Robotics Society on Advances In Robotics (AIR) <https://doi.org/10.1145/3610419.3610451>
- SCRAMPPI: Contingency-Constrained Planning using Hamilton–Jacobi Reachability*, submitted to ICRA 2027 - [arXiv](#)
- Implementation of a mobile robotic serial manipulator inside the stacks of vertical farms (*Patents in progress* - 202441084733)